ABDUL ALI

B.Tech Computer Science Student | Smart India Hackathon Grand Finalist

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PROFESSIONAL SUMMARY

Computer Science undergraduate and Smart India Hackathon Grand Finalist with hands-on experience across robotics, IoT, and machine learning. Skilled in building end-to-end ML/NLP solutions and full-stack web applications (Flask, JSP, JDBC, MySQL). Combines strong technical execution with proven experience teaching and mentoring others in applied technology.

EDUCATION

Bachelor of Technology, Computer Science and Engineering — *2023 – 2027 (Expected)*
Integral University • CGPA: 7.80

Intermediate — Physics, Chemistry & Mathematics — *2022 – 2023*
Safa Public School • 72%

High School — *2020 – 2021*
The Millennium School • 85%

PROFESSIONAL EXPERIENCE

Software Engineering Intern — Tata Consultancy Services (TCS) — *Jun 2026 – Aug 2026*

- Built a Course Difficulty Prediction Model using machine learning and NLP to analyze course content and predict difficulty levels.
- Applied NLP techniques for text preprocessing and feature extraction, and trained ML models to classify and score course difficulty.
- Developed and deployed a full-stack web application to serve the model, using Flask for the backend/API layer and JSP for the frontend.
- Integrated the trained ML model with the Flask API to enable real-time difficulty predictions within the application.
- Established a JDBC connection between the JSP application and a MySQL database, building data-fetching workflows to query course records and display real-time results.

Lead Trainer — Embedded Systems & IoT Workshop — Softflew Technologies — *Oct 2025 – Dec 2025*

- Mentored students and peers through hands-on training in embedded systems design and IoT architecture.
- Designed a practical curriculum covering microcontrollers, sensor integration, communication protocols, and IoT protocols.
- Guided participants in building real-world projects, including cloud telemetry, remote actuator control, and automated sensor node deployment.
- Facilitated interactive debugging sessions covering circuit troubleshooting, hardware-software co-design, and firmware best practices.

Head Faculty — Brain Wizard Summer Camp Workshop — *May 2025 – Jul 2025*

- Led hands-on robotics and IoT training sessions for students, designing and demonstrating real-time hardware-based projects.
- Taught foundational technical concepts to build creativity and practical understanding among participants.

PROJECTS

Smart Home Automation System with IoT | *ESP32, C++, Node.js, Python, Angular, React, MySQL*

- Programmed an ESP32 microcontroller in C++ to read sensor data and trigger relays for real-time remote appliance control.
- Developed high-throughput REST APIs in Node.js and Python to process telemetry data and stream device states.
- Designed a MySQL schema to log historical sensor readings, user preferences, and energy consumption metrics.
- Built dynamic web dashboards in Angular and React for real-time monitoring, scheduling, and power analytics.
- Integrated voice control and an automated energy-monitoring engine to reduce power waste.

Intelligent Pesticide Sprinkling System | ***Smart India Hackathon 2025***

- Developing an automated, sensor-driven pesticide sprayer for precision agriculture to optimize usage, reduce human exposure, and prevent wastage; designed to scale to any moving platform such as a drone or cart.
- Proposed by the Ministry of Education for deployment in the Punjab region.

AI/ML-Powered Institutional Performance Tracking System

- Creating a data-driven platform that leverages machine learning to analyze and enhance institutional performance metrics.

Personal Portfolio Website | *Angular*

- Built a personal portfolio website from scratch using Angular to showcase projects, skills, and experience.
- Designed a responsive, component-based UI with custom routing and animations for a smooth, modern browsing experience.

Additional Robotics & IoT Projects

- Obstacle Avoiding Car — Built and taught an autonomous car using ultrasonic sensors and Arduino to detect and avoid obstacles.
- Smart Doorbell — Developed an IoT-based doorbell system for remote alerts and monitoring.
- IoT-Based Dustbin — Created a sensor-enabled dustbin using NodeMCU to automate lid operation and monitor fill level.
- Touch-Activated Lamp — Designed a capacitive touch-controlled lamp to demonstrate interactive electronics.
- Maze-Solving Bot — Built a maze-solving robot using IR sensors and motor drivers, introducing logic-based navigation.
- Motion Detecting Device — Constructed a PIR-based motion detector to trigger security alerts.

Student Mentor | *Leadership & Mentorship*

- Provide ongoing career and technical guidance to help peers clarify academic goals and technical roadmaps.
- Onboard students into active engineering and software projects through targeted sub-tasks and hands-on skill-building sessions.

TECHNICAL SKILLS

- Machine Learning & NLP • MySQL • Web Development (HTML, CSS, JavaScript, Flask, JSP, JDBC, Angular)
- IoT & Embedded Systems • Microcontrollers • PCB Design • Electrical Components • Soldering
- Leadership & Mentoring

REFERENCES

Mohd Zaid
Dr. Mohammad Haroon
Shubhendra Yadav